

Weekly Notice 8

Exercises, week 8:

1. Exercise 78.
2. Exercise 82. Are the results (1) and (2) true also for extended stopping times?
3. Exercise 83.
4. Exercise 84(2)-(3).
5. Exercise 98. Brownian motions are defined in Definition 10.1 page 83.
6. Exercise 69.
7. Exercise 70.

Remark Let $(Y_t)_{t \geq 0}$ be progressively measurable with state space S . Let S' be another state space, $F : S \rightarrow S'$ be measurable, and define $Z_t = F(Y_t)$ for $t \geq 0$. Then $(Z_t)_{t \geq 0}$ is progressively measurable as well.

Proof: To show that $(Z_t)_{t \geq 0}$ is progressively measurable, we must show that for every $t \geq 0$, the map

$$\phi_t^Z : [0, t] \times \Omega \rightarrow S', \quad (s, \omega) \mapsto Z_s(\omega),$$

is $(\mathcal{B}([0, t]) \times \mathcal{F}_t, \mathcal{B}(S'))$ -measurable.

Fix $t \geq 0$. Since $(Y_t)_{t \geq 0}$ is progressively measurable, the map

$$\phi_t^Y : [0, t] \times \Omega \rightarrow S, \quad (s, \omega) \mapsto Y_s(\omega),$$

is $(\mathcal{B}([0, t]) \times \mathcal{F}_t, \mathcal{B}(S))$ -measurable. Since

$$Z_s(\omega) = F(Y_s(\omega)),$$

we have

$$\phi_t^Z = F \circ \phi_t^Y.$$

Therefore ϕ_t^Z is $(\mathcal{B}([0, t]) \times \mathcal{F}_t, \mathcal{B}(S'))$ -measurable as the composition of the measurable maps ϕ_t^Y and F .

As this holds for every $t \geq 0$, the process $(Z_t)_{t \geq 0}$ is progressively measurable.

Second mandatory assignment: See next page.

- (1) Let $(B_t)_{t \geq 0}$ denote a standard Brownian motion.
- (a) Determine the distribution of $B_s + B_t$ for $0 \leq s \leq t$. Determine the distribution of B_2^2 and of $B_1^2 + (B_2 - B_1)^2$.
 - (b) Argue that $\mathbb{P}(0 < B_1 < B_2 < B_\pi) = \frac{1}{8}$. Determine $\mathbb{E}[B_1 B_2 B_\pi | \mathcal{F}_2^B]$ and $\mathbb{E}[B_1 B_2 B_\pi]$.
 - (c) Define $U_t = tB_{\frac{1}{t}}$ for $t > 0$. Let $0 < s < t$. Determine the distribution of U_s and of $U_t - U_s$, and determine whether U_s and $U_t - U_s$ are independent.

For $a \in \mathbb{R}$ let

$$T_a = \inf\{t \geq 0 : B_t = a\}.$$

Later in the course we show both that $\mathbb{P}(T_a < \infty) = 1$ for all $a \in \mathbb{R}$ and that $\mathbb{P}(T_a < T_b) = b/(b - a)$ for $a < 0 < b$.

- (d) Determine $\mathbb{P}(T_1 < T_{-1} < T_2)$.
- (e) Define the process $(X_t)_{t \geq 0}$ by

$$X_t = B_{2t} - B_t, \quad t \geq 0.$$

Does

$$X_t \stackrel{d}{=} B_t \quad \text{hold for all } t \geq 0?$$

Is $(X_t)_{t \geq 0}$ a Brownian motion?

- (f) Let $Y \sim N(0, 1)$ and assume that Y is independent of $(B_t)_{t \geq 0}$. Let

$$Z_t = B_t + t(Y - B_1), \quad t \in [0, 1].$$

Are $(B_t)_{t \in [0, 1]}$ and $(Z_t)_{t \in [0, 1]}$ versions? Are they modifications or indistinguishable (dansk: uskelnelige)?

- (2) Let $(N_t)_{t \geq 0}$ denote a Poisson process with parameter $\lambda > 0$.
- (a) Determine $\mathbb{E}[T_{12}]$ and $\text{Var}(T_{12})$. Also determine the distribution of $T_8 - T_5$.
 - (b) Determine $\mathbb{E}[T_{11} | \{N_2 = 5\} \cap \{T_3 < 2\}]$.
 - (c) Determine $\mathbb{E}[N_5 | N_2 = 7]$ and $\text{Var}[N_5 | N_2 = 7]$. Also determine $\mathbb{E}[N_5 | N_2]$.

- (3) Durrett, Exercise 2.27.

The Reference to Durrett is *Essentials of Stochastic Processes* (2016 edition) which you can download from library.au.dk.